MZUZU UNIVERSITY

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FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4703: NETWORK ADMINISTRATION AND INFORMATION SECURITY

SUPPLEMENTARY EXAMINATION

DATE: December,2021 **TIME ALLOWED: 2:30 HOURS**

INSTRUCTIONS

i. There are **5** questions in this paper. **ANSWER ALL.**

i. There are **TWO** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

iii. Please begin each answer on a fresh page and indicate the question number on each page

iii. Write your answers clearly.

Question 1

a) Define the following concepts:

i. Network security. **[2 Marks]**

ii. Authentication. **[2 Marks]**

iii. Encryption. **[2 Marks]**

b) Explain the **three** concepts of the CIA triad and for each give an example of how you would achieve this in securing your network. **[6 Marks]**

c) Discuss the **three** components of an intrusion detection system. **[6 Marks]**

d) Explain **one** challenge of an intrusion detection system. **[2 Marks]**

Question 2

a) Explain how a Virtual Private Network(VPN) works. **[4 Marks]**

b) Discuss the concept of One Time Passwords (OTP) and how they are used to authenticate users accessing critical applications. In your explanation include **two** advantages of incorporating OTP into financial applications. **[6 Marks]**

c) Explain the concept of multi-factor authentication and give an example of how you could implement multi-factor authentication. **[4 Marks]**

d) What is a computer Virus? **[2 Marks]**

e) Mention any 4 indicators of a computer being infected by a virus. **[4 Marks]**

Question 3

a) What is a Certificate Revocation List? **[2 Marks]**

b) Explain two reasons why a Certificate may be revoked? **[4 marks]**

c) Explain the difference between symmetric and asymmetric encryption. **[2 Marks]**

d) Mention and explain two wireless network threats that you know. **[4 Marks]**

e) Explain three mechanisms you would use to implement authentication and for each mechanism give an example. **[6 Marks]**

f) Explain why should you constantly audit your wireless network? **[2 Marks]**

Question 4

a) Explain **three** characteristics of a biometric system. **[6 Marks]**

b) Explain what happens in a denial of service attack. **[4 Marks]**

c) Network controls are categorized into **three**, discuss network controls and give one example on each control. **[6 Marks]**

d) Explain how you would prevent Phishing attacks within your networks? **[4 marks]**

Question 5

a) Discuss how a firewall works and its benefits to an organization. **[4 Marks]**

b) Define a security bug then explain how it is a threat to our networks. **[4 Marks]**

c) Discuss an SQL injection attack. **[4 Marks]**

d) Explain the password rule of thumb. **[6 Marks]**

e) What is War dialing attack? **[2 Marks]**

END OF QUESTION PAPER!